

Interim Report — Portfolio Optimization (Week 9)

Guide Me in Finance (GMF) Investments Author: Yostina Abera · bet30539@gmail.com **Scope:** Task 1 — Data preprocessing, exploratory analysis, stationarity testing, and foundational risk metrics **Assets:** TSLA (high-growth equity) · SPY (broad market) · BND (investment-grade bonds) **Period:** 2015-01-01 → 2026-06-30 · **Source:** Yahoo Finance ([yfinance](#)) **Supporting notebook:** [notebooks/01_task1_eda.ipynb](#)

1 · Data Extraction & Cleaning

Extraction. Daily OHLCV data plus dividend/split-adjusted close was pulled for TSLA, BND, and SPY via [yfinance](#) (script: [scripts/fetch_data.py](#)). The downloader keeps both the raw **Close** and **Adjusted Close** (`auto_adjust=False`), retries with exponential back-off against rate limits, and writes two artifacts:

- [data/processed/prices.csv](#) — tidy long format (Date, Ticker, OHLCV, Adj Close)
- [data/processed/adj_close.csv](#) — wide adjusted-close matrix (dates × tickers) for modeling

All return-based analysis uses **Adjusted Close**, the correct basis for total-return measurement.

Data-quality assessment and handling.

Check	Finding	Action
Coverage	2,889 trading days per asset (2015-01-02 → 2026-06-30)	None needed
Data types	Dates parsed to <code>datetime64</code> ; prices <code>float64</code> ; volume <code>int</code>	Enforced on load
Missing values	Zero nulls in any column	None needed
Calendar alignment	All three trade on the identical NYSE calendar — perfectly aligned	Verified
Robustness guard	—	Reindexed onto the full business-day calendar + forward-fill, so the pipeline stays correct if the data source changes

Summary: the dataset was **clean on arrival** — no missing values, no type coercion required, and full calendar alignment across the three assets. A defensive reindex/forward-fill step is retained for reproducibility. Daily simple returns were engineered as the primary analysis unit (scale-free and far closer to stationary than raw prices).

2 · Key EDA Visualizations & Insights

Three visualizations carry the core story (all reproduced in the notebook):

① **Growth of \$1 invested (normalized prices).** Placing all assets on a common base exposes relative performance:

Asset	\$1 in 2015 → June 2026
TSLA	\$28.77
SPY	\$4.40
BND	\$1.23

Insight: Tesla delivered ~29× growth — an order of magnitude above the market — but, as the volatility analysis shows, at dramatically higher risk. SPY compounded steadily; BND barely moved, consistent with its role as a capital-preservation holding.

② **Daily returns time-series + distributions.** The return series and histograms make the risk ordering visually unambiguous: TSLA's spread dwarfs SPY's, which dwarfs BND's. All three distributions are **fat-tailed** (excess kurtosis) — extreme days occur far more often than a normal distribution predicts, which is why we report *historical* VaR alongside the parametric estimate.

③ **Rolling 21-day volatility.** Rolling standard deviation reveals strong **volatility clustering** — calm regimes punctuated by synchronized spikes across all assets, most dramatically the **COVID crash (Feb–Mar 2020)**, which lifted volatility even in the normally-placid bond fund.

Outliers / extreme days ($|z| > 3\sigma$): TSLA 50 · SPY 42 · BND 35. The largest single-day moves map to identifiable market events:

Asset	Best day	Worst day
TSLA	+22.7% (2025-04-09)	−21.1% (2020-09-08)
SPY	+10.5% (2025-04-09)	−10.9% (2020-03-16)
BND	+4.2% (2020-03-13)	−5.4% (2020-03-12)

Insight: BND records the **fewest** z-score outliers and its "extremes" are sub-1% in absolute terms; TSLA's outliers are genuine double-digit swings driven by earnings/product news. The shared April-2025 and March-2020 dates confirm system-wide risk events rather than idiosyncratic noise.

3 · Stationarity Test Results & Interpretation

Augmented Dickey-Fuller (ADF) test — H_0 : unit root (non-stationary); reject at $p < 0.05$.

Series	ADF p-value (price level)	ADF p-value (daily return)
TSLA	0.746 → non-stationary	≈ 0 → stationary
SPY	0.997 → non-stationary	≈ 0 → stationary
BND	0.710 → non-stationary	≈ 0 → stationary

Interpretation. Every raw price series fails to reject the unit-root null — prices trend and their variance grows over time, behaving like a **random walk** (consistent with the **Efficient Market Hypothesis**: past prices don't cleanly predict future ones). Differencing once — which is effectively what a daily return is — removes the trend and yields a stable mean/variance, so returns are strongly stationary.

Modeling implication. ARIMA requires a stationary input, so the price series needs **one order of differencing ($d = 1$)** — or, equivalently, we model returns directly. This single result is what justifies the "I" (integrated) term in ARIMA and directly informs Task 2.

4 · Volatility Analysis & Risk Metrics

Annualized risk/return profile (daily stats annualized: return $\times 252$, volatility $\times \sqrt{252}$):

Asset	Ann. return	Ann. volatility	Skew	Excess kurtosis	Sharpe (rf = 2%)
TSLA	45.6%	57.2%	+0.29	4.5	0.76
SPY	14.5%	17.7%	−0.31	14.0	0.71
BND	2.0%	5.3%	−0.92	36.5	≈ 0.00

Value at Risk (VaR) — maximum expected single-day loss, reported as a positive % loss:

Asset	Historical VaR 95%	Historical VaR 99%	Parametric VaR 99%
TSLA	5.17%	8.99%	8.20%
SPY	1.66%	3.17%	2.53%
BND	0.48%	0.86%	0.77%

Insights.

- **Risk is cleanly ranked:** TSLA » SPY » BND on both volatility and VaR. On a bad 1-in-20 day, TSLA can lose ~5% versus BND's ~0.5% — roughly an order of magnitude apart.
- **Fat tails are real:** for every asset **historical VaR exceeds parametric (Gaussian) VaR** at the 99% level, confirming the non-normality seen in the return histograms — the normal model *understates* extreme losses. High excess kurtosis (SPY 14, BND 36.5) quantifies this.
- **Risk-adjusted return:** over this 2015–2026 window **TSLA posts both the highest raw return and the highest Sharpe (~0.76)**, narrowly ahead of SPY (~0.71) — its outsized returns compensated for its volatility in this sample. SPY earns nearly the same risk-adjusted return with a fraction of the risk (the more robust core), while **BND's Sharpe is ≈ 0** (its return barely clears the 2% risk-free rate) — its job is stability, not return.
- **Volatility clustering** (rolling-std analysis) means variance is time-varying, not constant — a reason to treat any single volatility estimate with caution.

5 · Conclusion & Next Steps

The three assets are well-chosen complements: **TSLA** supplies return potential, **BND** supplies ballast, and **SPY** supplies diversified core exposure. Their differing risk profiles and imperfect correlations are exactly

the raw material needed for portfolio optimization.

The data is clean and modeling-ready. The stationarity results ($d = 1$) directly configure the forecasting models in **Task 2 (ARIMA/SARIMA vs. LSTM)**, and the risk metrics established here (VaR, volatility, Sharpe) provide the baseline against which the optimized portfolio will be evaluated in **Task 4 (Efficient Frontier & backtesting)**.

Key caveat carried forward: the random-walk / EMH evidence warns that **pure long-horizon price prediction is unreliable** — so downstream decisions should lean on diversification and optimization rather than point-price forecasts.